

A gcd-sum evaluation of a totient-like function: a proof of Schulte’s conjecture on OEIS A358319

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Abstract

Let b be the multiplicative function determined by $b(p^e) = ((p-2) - (p-1)e)p^{e-1}$ for $e \geq 1$. In November 2022 Werner Schulte conjectured on OEIS A358319 that $b(n) = \sum_{k=1}^n \gcd(k, n) \mu(\text{rad}(\gcd(k, n)))$ for all $n \geq 1$, where $\mu(\text{rad}(m)) = (-1)^{\omega(m)}$ is OEIS A076479. We prove it. The argument groups the summation range by the value of $\gcd(k, n)$, which converts the sum into a Dirichlet convolution with Euler’s totient, and then evaluates the local factor at a prime power, where all but two of the terms are equal and collapse into a single multiple of $p^e - p^{e-1}$. The result is elementary; its interest is that the statement has stood labelled as a conjecture on the entry since 2022.

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1 The sequence and the conjecture

Throughout, p denotes a prime, φ is Euler’s totient function (OEIS A000010), $\omega(m)$ is the number of distinct prime factors of m , and $\text{rad}(m)$ is the product of those primes. We write

$$\chi(m) = \mu(\text{rad}(m)) = (-1)^{\omega(m)},$$

which is OEIS A076479; it is multiplicative, with $\chi(1) = 1$ and $\chi(p^j) = -1$ for every $j \geq 1$. Note that χ is not completely multiplicative and is not the Möbius function: it is -1 at every prime power, including those that are not squarefree.

OEIS A358319 is defined as the multiplicative function

$$b(p^e) = ((p-2) - (p-1)e)p^{e-1} \quad (p \text{ prime, } e \geq 1), \tag{1}$$

extended by $b(1) = 1$ and $b(n) = \prod_{p^e \parallel n} b(p^e)$. The entry has offset 1 and begins

$$b(1), \dots, b(12) = 1, -1, -1, -4, -1, 1, -1, -12, -9, 1, -1, 4.$$

For $e = 1$ the local value (1) is $(p-2) - (p-1) = -1$, which is why b takes the value -1 at every prime.

The entry carries the following line in its FORMULA section:

Conjecture: $a(n) = \text{Sum}_{k=1..n} \gcd(k, n) * \text{A076479}(\gcd(k, n))$ for $n > 0$.

The conjecture is due to Werner Schulte, who created the entry on 9 November 2022. As of the “Last modified” line on the live entry (30 November 2025, revision 11) the statement is still recorded as a conjecture; the only link on the entry is a table of values by Andrew Howroyd, and no proof appears among the entry’s links, formulas or programs.

Write

$$T(n) = \sum_{k=1}^n \gcd(k, n) \chi(\gcd(k, n)). \tag{2}$$

The claim to be proved is that $T(n) = b(n)$ for every $n \geq 1$.

2 The gcd-sum lemma

The summand in (2) depends on k only through $\gcd(k, n)$, so the sum can be collapsed by grouping the summation range according to that value.

Lemma 1. *Let f be any arithmetic function and let $n \geq 1$. Then*

$$\sum_{k=1}^n f(\gcd(k, n)) = \sum_{d|n} f(d) \varphi(n/d) = (f * \varphi)(n),$$

where $*$ denotes Dirichlet convolution.

Proof. Every k with $1 \leq k \leq n$ has $\gcd(k, n) = d$ for exactly one divisor d of n , so

$$\sum_{k=1}^n f(\gcd(k, n)) = \sum_{d|n} f(d) \cdot \#\{k : 1 \leq k \leq n, \gcd(k, n) = d\}.$$

Fix a divisor d and put $k = dm$. Then $1 \leq k \leq n$ becomes $1 \leq m \leq n/d$, and $\gcd(k, n) = d$ becomes $\gcd(m, n/d) = 1$, so the inner count is $\varphi(n/d)$. As d runs over the divisors of n so does n/d , and the sum is the convolution $(f * \varphi)(n)$. $\square$

Applying Lemma 1 to (2) with $f(d) = d\chi(d)$ gives

$$T = f * \varphi, \quad f(d) = d\chi(d), \quad f(p^j) = -p^j \ (j \geq 1), \quad f(1) = 1. \quad (3)$$

Here f is multiplicative, being the product of the multiplicative functions $d \mapsto d$ and χ , and φ is multiplicative; hence T is multiplicative and it suffices to compare T with b at a single prime power.

3 The theorem

Theorem 2. *For every $n \geq 1$,*

$$\sum_{k=1}^n \gcd(k, n) \mu(\text{rad}(\gcd(k, n))) = b(n),$$

where b is the multiplicative function defined by (1). That is, Schulte's conjecture on OEIS A358319 is true.

Proof. By (3) both T and b are multiplicative, so it is enough to prove $T(p^e) = b(p^e)$ for every prime p and every $e \geq 1$.

Expand the convolution and separate the two extreme terms, $j = 0$ and $j = e$, from the rest:

$$T(p^e) = \sum_{j=0}^e f(p^j) \varphi(p^{e-j}) = \underbrace{\varphi(p^e)}_{j=0} + \sum_{j=1}^{e-1} (-p^j)(p^{e-j} - p^{e-j-1}) + \underbrace{(-p^e)}_{j=e}.$$

The extreme terms are separated because $f(p^0) = 1$ rather than -1 , and because $\varphi(p^0) = 1$ rather than $p^0 - p^{-1}$; every term strictly between them has the same shape. Indeed, for $1 \leq j \leq e-1$,

$$p^j(p^{e-j} - p^{e-j-1}) = p^e - p^{e-1},$$

independently of j , and there are exactly $e-1$ such terms. Writing $\varphi(p^e) = p^e - p^{e-1}$ as well, we get

$$T(p^e) = (p^e - p^{e-1}) - (e-1)(p^e - p^{e-1}) - p^e = (2-e)(p^e - p^{e-1}) - p^e. \quad (4)$$

Now factor out p^{e-1} from (4), using $p^e - p^{e-1} = p^{e-1}(p - 1)$ and $p^e = p^{e-1} \cdot p$:

$$T(p^e) = p^{e-1} \left[(2 - e)(p - 1) - p \right].$$

Finally expand the bracket:

$$(2 - e)(p - 1) - p = 2p - 2 - e(p - 1) - p = (p - 2) - (p - 1)e.$$

Hence $T(p^e) = ((p - 2) - (p - 1)e)p^{e-1} = b(p^e)$ by (1), and multiplicativity finishes the proof. $\square$

Remark 3. The proof accounts for the shape of (1). The linear dependence on the exponent e , which is the only unusual feature of the local value, comes from the $e - 1$ interior terms of the convolution being identically equal to $-(p^e - p^{e-1})$; the constant $(p - 2)$ is what survives from the two extreme terms once that multiple is removed. It also settles the sign. The bracket $(p - 2) - (p - 1)e$ is strictly decreasing in e and already equals -1 at $e = 1$, so $b(p^e) < 0$ for every prime p and every $e \geq 1$; it vanishes only where $e = (p - 2)/(p - 1)$, which lies in $[0, 1)$ and is therefore never an admissible exponent. Multiplicativity then gives $\text{sign } b(n) = (-1)^{\omega(n)}$ for all $n \geq 1$, and b is nowhere zero.

4 Verification

The identity was checked numerically by two deliberately different routes, neither of which was used in the proof.

The first computes the right-hand side of (2) directly, forming $\text{gcd}(k, n)$ for each $k = 1, \dots, n$ and evaluating χ on it by factorisation. The second builds $b(n)$ from the multiplicative definition (1) alone, factoring n and multiplying the local values; it never forms a gcd. The two agree with each other for all n with $1 \leq n \leq 400$, and each agrees with all 80 terms of the DATA section published on the entry, over its full range $n = 1, \dots, 80$.

Separately, the local identity $T(p^e) = b(p^e)$ established in the proof was confirmed directly from the convolution $\sum_{d|p^e} f(d)\varphi(p^e/d)$, without the collapse (4), for $p \in \{2, 3, 5, 7\}$ and $1 \leq e \leq 5$.

5 Concluding remarks

The result is elementary, and the entry already stands one step from it: A358319 carries the unlabelled formula “Equals Dirichlet convolution of A000010 and $n \cdot \text{A076479}$ ”, which is precisely the statement $b = f * \varphi$ of (3). Given that line, the conjecture follows from Lemma 1 alone. As with the companion entry, the substance of this note is that the entry records the same identity twice, once as a conjecture and once as a fact, and that neither record carries a proof.

The companion entry OEIS A358272, also due to Schulte, carries the conjecture $a(n) = \sum_{k=1}^n \text{gcd}(k, n)\lambda(\text{gcd}(k, n))$ for the multiplicative function with $a(p^e) = (-1)^e p^{2\lfloor e/2 \rfloor}$, where λ is the Liouville function. It is proved by the same lemma with a different local computation, in which an alternating sum collapses by the parity of e ; that case is treated separately. The two conjectures should be regarded as one piece of mathematics applied twice, not as independent results.

A third conjecture of the same family, OEIS A384531, is already settled: the entry links a note of David G. Radcliffe of June 2025 proving $\sum_{i=1}^n \text{gcd}(i, n)\text{gcd}(i + 1, n)$ multiplicative with the stated local values, so despite the label still carried there, that statement is not open.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A358319, formulas of W. Schulte, 9 November 2022; sequences A000010, A076479, A358272.
- [2] D. G. Radcliffe, *Notes on two arithmetical functions*, June 2025; linked from OEIS A384531.